

1. There exist two processes P_1 and P_2 in a multiprogramming batch system. P_2 enters into the system 10ms later than P_1 , and their executing traces, i.e. alternating sequences of CPU bursts and I/O bursts, are as follows:

P_1 : computing, 80ms $\rightarrow$ I/O operation, 100ms $\rightarrow$ computing, 40ms

P_2 : computing, 130ms $\rightarrow$ I/O operation, 50ms $\rightarrow$ computing, 50ms

It is assumed that time costs of CPU scheduling and process switch are omitted, what is the maximal throughput for completing these two processes, and why?

2. Consider the following set of processes P_1 , P_2 , P_3 and P_4 . For $1 \leq i \leq 4$, the arrival time of each P_i , the length of the CPU burst time of each process P_i , and the priority number for each P_i are given as below, and a smaller priority number implies a higher priority.

Process	Arrival Time	Burst Time	Priority Number
P_1	0.00	4.001	2
P_2	3.00	3.001	1
P_3	4.00	3.001	4
P_4	5.00	5.001	3

For a process having been in the system, its *response-ratio* is defined as its waiting time divided by its CPU burst time (i.e. waiting time/CPU burst time). When the CPU is idle, Highest Response-Ratio (HRR) scheduling selects the process with the highest response-ratio to allocate CPU, and a process in the waiting state is not allowed to preempt the CPU allocated to the other process in the running state. Suppose that HRR scheduling is employed.

- 1) Draw a Gantt chart illustrating the execution of these processes
- 2) What are the average waiting time and the average turnaround time

3. In a kindergarten, the teacher distributed the fruits to children. All fruit would be placed on a large plate. The boys liked eating apples and the girls liked oranges. At the beginning, the plate was empty, which could contain 5 apples or oranges altogether. The teacher would put either an apple or two oranges into the plate each time. When she put an apple in the plate, a boy would get and eat it; and when she put two oranges in, a girl would get the two oranges and eat them.

- 1) Please design appropriate semaphore-based processes to describe the behaviors of the teacher and boys and girls.
- 2) If the teacher will put three oranges into the plate instead of two each time, and a girl will still get two oranges each time, modify your program for the new conditions.

It is required that

- 1) definition and initial value of each semaphore, and
- 2) the algorithm to coordinate the production process for the three workers should be given.

4 Can the following algorithm satisfy all three requirements for the critical-section problem? If yes, please prove; if no, why?

```
boolean flag[2] = {false, false};
```

```
int turn=0;
```

```
process P0
```

```
do{
```

```
    flag[0] = true;           (1)
```

```
    while(turn!=0){          (2)
```

```
        while(flag[1]);      (3)
```

```
        turn=0;              (4)
```

```
    }
```

```
    critical section          (5)
```

```
    flag[0] = false;
```

```
    remainder section
```

```
}while(TRUE)
```

```
process P1
```

```
do{
```

```
    flag[1]=true;            (6)
```

```
    while(turn!=1){          (7)
```

```
        while(flag[0]);      (8)
```

```
        turn=1;              (9)
```

```
    }
```

```
    critical section          (10)
```

```
    flag[1]=false;
```

```
    remainder section
```

```
}while(TRUE)
```